

SHIVESH SARAVANAN

☎ +1 817-723-9353 | ✉ sh4ivu@gmail.com | 📍 Arlington, TX | [in shivesh-s](#) | [Shivesh4](#) | [Portfolio](#)

EDUCATION

The University of Texas at Arlington

Aug. 2024 - May 2026

Master of Science in Computer Science

Relevant Coursework: Algorithms, Advanced Software Engineering, Machine Learning, Artificial Intelligence I

Amrita Vishwa Vidyapeetham

Sep. 2020 - July 2024

Bachelor of Technology in Computer Science and Engineering

Relevant Coursework: Algorithms (I & II), Distributed Systems, Computer Networks

TECHNICAL SKILLS

Languages: C, C++, CSS, Golang, Haskell, HTML, Java, JavaScript, Python, Scala, Solidity, SQL

Frameworks & Libraries: Angular, Bootstrap, Flask, Hadoop, Hive, JQuery, NLTK, Node.js, NumPy, pandas, Pig, PyTorch, React, Sklearn, Spark, TensorFlow, Truffle, Web3.js

Developer Tools: Amazon Web Services, Arduino, Cisco Packet Tracer, Docker, Eclipse, Git, Jenkins, Jira, Kubernetes, Metamask, Selenium, Visual Studio, Wireshark

Database: Cassandra, DynamoDB, MongoDB, MySQL

Operating System: Linux, macOS, Windows

Methodologies: Agile (Scrum, CI/CD), Automation Testing, Ensemble Learning, System Design, Transfer Learning

EXPERIENCE

Software Developer

Aug. 2023 – July 2024

Holistic MD LLP

Coimbatore, India

- Engineered Angular SPAs with **HttpClient** and **RxJS** to cut API calls and improve load times by **32%**, while enabling **10,000+** concurrent WebSocket sessions with **1.3× faster throughput** and **35%** lower resource usage.
- Containerized 6+ services with **Docker** to streamline setup by **50%** and cut bugs by **30%**, and integrated **Spring Boot** microservices with authentication, CORS, and error handling.
- Integrated Stripe payments and collaborated with Flutter team to ensure API compatibility and consistent cross-platform UX, while enabling **CI/CD** pipelines with **Git + Jenkins** for Dockerized deployments

Machine Learning Engineer Intern

Nov. 2022 – March 2023

TIFAC-CORE in Cyber Security

Coimbatore, India

- Automated reverse-engineering and emulated a WhatsApp GIF-based RCE on legacy builds in isolated environments, reducing analysis time by **45%**
- Developed **PCAP-based** data pipelines for network forensics and traffic classification, integrating ML insights into reports for **10+ systems** to accelerate patch deployment
- Designed a Neural Network model to distinguish encrypted vs. compressed traffic with **74% accuracy**

PROJECTS

Peer Tutoring & Study Planner | *React, Node.js, MongoDB, WebSockets, AWS*

- Designed a modular platform with dedicated student/tutor dashboards, role-based routing, and secure access flows, supporting **5,000+ users** with smooth real-time interactions.
- Built an **AI-driven Tutorbot** using OpenAI LLMs to deliver personalized study plans and ranked tutor suggestions, improving learning engagement by **40%**.
- Created a 4-week study planner with adaptive tasks, progress tracking, and targeted announcements to increase platform activity by **30%**.

AI-Powered Career Guidance Platform | *React, Express.js, PostgreSQL, Docker, AWS*

- Architected a scalable career platform with job search, resume uploads, profile management, and application tracking, designed to handle **10,000+ job listings** and **800+ concurrent users**
- Designed modular microservices with **PostgreSQL**, **REST APIs**, and **Docker** on **AWS** to cut response times by **30%** and enable horizontal scaling
- Integrated **AI-driven skill gap analysis** and **resume feedback** using OpenAI LLMs to boost job match accuracy by **30%** and user engagement by **45%**

Intelligent Expense Management System | *Flask, MySQL, LSTM, NLP, OCR*

- Designed an expense platform with Flask and MySQL using LSTM forecasting to achieve **92%** prediction accuracy
- Created a **recommendation engine** using collaborative filtering and classification to improve purchase accuracy
- Automated bill processing with **OCR** and **NLP** to cut entry time by **80%** and achieve **98%** accuracy in labeling